```
import pandas as pd
import numpy as np
df=pd.read_csv('/content/insurance.csv')
```

```
df.head()
```

	age	sex	bmi	children	smoker	region	charges
0	19	female	27.900	0	yes	southwest	16884.92400
1	18	male	33.770	1	no	southeast	1725.55230
2	28	male	33.000	3	no	southeast	4449.46200
3	33	male	22.705	0	no	northwest	21984.47061
4	32	male	28.880	0	no	northwest	3866.85520

```
df.columns
```

```
Index(['age', 'sex', 'bmi', 'children', 'smoker', 'region', 'charges'], dtype='object')
```

```
df.isnull().sum()
```

	0
age	0
sex	0
bmi	0
children	0
smoker	0
region	0
charges	0

dtype: int64

```
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
```

```
from sklearn.preprocessing import LabelEncoder
```

```
le = LabelEncoder()

df['sex'] = le.fit_transform(df['sex'])
df['smoker'] = le.fit_transform(df['smoker'])
df['region'] = le.fit_transform(df['region'])
```

```
X = df[['age','bmi','children','smoker']]
y = df['charges']
```

```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
```

```
model = LinearRegression()
model.fit(X_train, y_train)
```

```
▾ LinearRegression
LinearRegression()
```

```
y_pred = model.predict(X_test)
```

```
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score
```

```
mae = mean_absolute_error(y_test, y_pred)
print("MAE:", mae)
```

```
MAE: 4213.798594527248
```

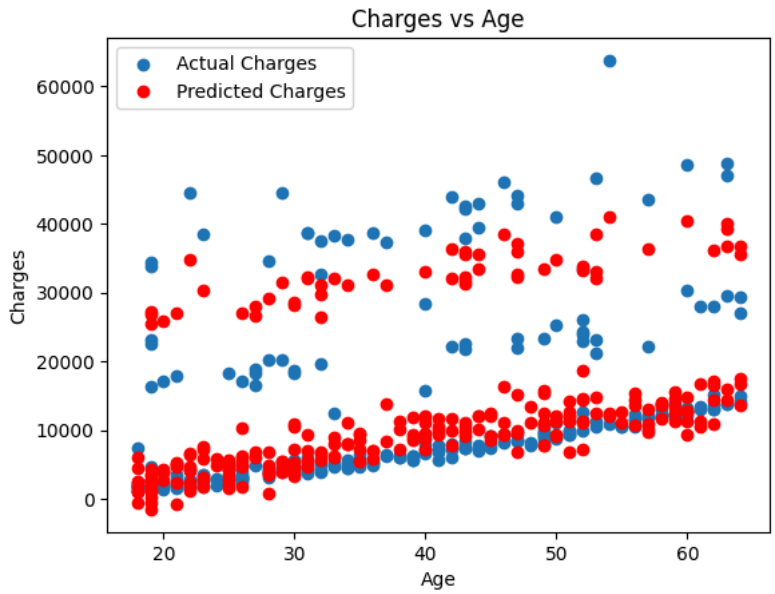
```
mse = mean_squared_error(y_test, y_pred)
print("MSE:", mse)
```

```
MSE: 33981653.95019776
```

```
r2 = r2_score(y_test, y_pred)
print("R2 Score:", r2)
```

```
R2 Score: 0.7811147722517886
```

```
coefficients = pd.DataFrame(model.coef_, X.columns, columns=['Coefficient'])
print(coefficients)
```

```
           Coefficient
age         257.071158
bmi         327.522631
children    427.199971
smoker    23653.211646
```

```
import matplotlib.pyplot as plt

plt.scatter(X_test['age'], y_test, label='Actual Charges')
plt.plot(X_test['age'], y_pred, 'o', color='red', label='Predicted Charges')
plt.xlabel("Age")
plt.ylabel("Charges")
plt.title("Charges vs Age")
plt.legend()
plt.show()
```